EduPlanBench: Benchmarking Education Agents as Long-Horizon Closed-Loop Planners

Supplementary Material

F Evaluation Metrics and Scoring

All metrics are calculated from recorded episode traces and then averaged across evaluated episodes. We distinguish (i) terminal mastery and task-aware success, (ii) shared trajectory-level performance, and (iii) track-specific educational diagnostics.

F.1 Shared Outcome and Cost Metrics

For episode n, let $\widehat{M}_0^{(n)}$ be the mean initial *visible* mastery over target concepts, $M_T^{(n)}$ the mean final hidden mastery, and $M_{\text{tar}}^{(n)}$ the mastery threshold. The public evaluator computes the episode-level progress and mastery-success indicators as

$$\begin{aligned} \text{PR}_n &= \text{clip}_{[0,1]}\left(\frac{M_T^{(n)} - \widehat{M}_0^{(n)}}{\max(10^{-6}, M_{\text{tar}}^{(n)} - \widehat{M}_0^{(n)})}\right), \\ \text{MG}_n &= \mathbb{I}[M_T^{(n)} \geq M_{\text{tar}}^{(n)}]. \end{aligned} \tag{S1}$$

Dataset-level PR and mastery GSR (MGSR) are the respective means of these episode quantities.

Let T_n be the executed step count, H_n the nominal task horizon, and v_t the action-validity indicator. Then

$$\text{Valid}_n = \frac{1}{T_n}\sum_{t<T_n} v_t, \qquad \text{NormSteps}_n = \frac{T_n}{\max(1, H_n)}. \tag{S2}$$

The context-cost proxy is the sum of serialized observation/action lengths, approximating each ASCII character by 1/4 token and each non-ASCII character by one token; it is not provider-billed token usage. The normalized context term is $\min(1, \text{Context}_n/10{,}000)$. Simulated learner time is the sum of elapsed-time fields in environment feedback.

F.2 Track-Aware Goal Success

Let G_n be the episode-level goal-success indicator. In all tracks, mastery attainment ($\text{MG}_n = 1$) is sufficient for $G_n = 1$. Otherwise, the public evaluator applies track-specific criteria, yielding

$$G_n = \begin{cases} \text{MG}_n \vee \left(Z_n^{\text{diag}} \wedge Z_n^{\text{rem}} \wedge Z_n^{\text{ground}} \wedge Z_n^{\text{adapt}}\right), & \text{Track 2}, \\ \text{MG}_n \vee \left(Z_n^{\text{path}} \wedge Z_n^{\text{prog}}\right), & \text{Track 1}, \\ \text{MG}_n, & \text{Track 3}, \end{cases} \tag{S3}$$

where, for the misconception track, $Z_n^{\text{diag}} = \mathbb{I}[\text{MisAcc}_n \geq 0.35]$, $Z_n^{\text{rem}} = \mathbb{I}[\text{RemSuccess}_n = 1]$, $Z_n^{\text{ground}} = \mathbb{I}[\text{FeedGround}_n \geq 0.35 \ \vee \ \text{RemMatch}_n \geq 0.05]$, and $Z_n^{\text{adapt}} = \mathbb{I}[\text{Adapt}_n \geq 0.30]$. For the MOOC track, Z_n^{path} indicates simultaneous satisfaction of $\text{PrereqViol}_n \leq 0.10$, $\text{ResMatch}_n \geq 0.50$, $\text{ConsSat}_n \geq 0.70$, and $\text{PathCoh}_n \geq 0.70$; $Z_n^{\text{prog}} = \mathbb{I}[\text{PR}_n \geq 0.50]$. GSR is $N^{-1}\sum_n G_n$. Thus GSR and MGSR are distinct, and GSR need not coincide with reaching the mastery threshold in Tracks 1 and 2.

F.3 Shared Core Score

In the implemented evaluator, adaptation quality is an *observable action-history proxy*, rather than a direct measurement of successful belief updating:

$$\text{Adapt}_n = \min\left(1, \frac{U_n + 0.5F_n^{\text{obs}}}{\max(1, T_n)}\right), \tag{S4}$$

where U_n counts actions with a nonempty plan-update field or an `update_plan` action, and F_n^{obs} counts observations containing recent feedback. The episode-level shared score is

$$\begin{aligned}\text{Core}_n = \text{clip}_{[0,1]}\big[&0.35G_n + 0.25\text{PR}_n + 0.15\text{Adapt}_n + 0.10\text{Valid}_n \\ &- 0.10\text{NormSteps}_n - 0.05\min(1, \text{Context}_n/10{,}000)\big].\end{aligned} \tag{S5}$$

F.4 Track-Specific Educational Diagnostics

Track 1: resource planning. For each step t, let C_t be the concepts addressed by the selected resource (or the action's declared targets), and $S_{<t} = \bigcup_{j<t} C_j$. Let P denote the required prerequisite set and $K^\star$ the target-concept set. The implemented prerequisite-violation count increments when C_t intersects $K^\star$, P is nonempty, and $P \cap S_{<t} = \varnothing$; the violation rate is this count divided by $\max(1, T)$. This is a *seen-prerequisite proxy*, not a proof that all prerequisites have been mastered. Resource Match is $T^{-1}\sum_t \mathbb{I}[C_t \neq \varnothing]$, while Path Coherence is $T^{-1}\sum_t \mathbb{I}[C_t \cap K^\star \neq \varnothing]$. Sequence Consistency penalizes both prerequisite violations and nonoverlapping switches of addressed concepts. Constraint Satisfaction subtracts a weighted combination of prerequisite violations, overload, invalid actions, and nominal-horizon excess:

$$\begin{aligned}\text{ConsSat} = \text{clip}_{[0,1]}\big[&1 - 0.45\text{PrereqViol} - 0.35\text{Overload} \\ &- 0.15(1 - \text{Valid}) - 0.05\text{OverBudget}\big].\end{aligned} \tag{S6}$$

Overload counts selected resources whose assigned difficulty exceeds their current visible mean mastery by more than 0.30; Plan Drift is $1 - \text{PathCoh}$. Difficulty Alignment averages $\max(0, 1 - |d_t - \min(\widehat{M}_t + 0.15, 0.8)|/0.8)$ over selected resources.

Track 2: diagnosis and remediation. Misconception Accuracy is the maximum reference-token recall of action diagnosis strings or rationales, using unique lowercase whitespace tokens of length greater than two; in reference-free cases, the implementation instead uses the presence of a diagnosis or explanation action. Remediation Match uses the same maximum token-overlap mechanism over concatenated misconception and remediation references and action rationales, plan updates, or payload text. By contrast, $\text{RemSuccess} = 1$ whenever at least one review, explanation, easier-problem, or similar-problem action is selected; it does *not* require a reference match or measured improvement. Feedback Grounding assigns a step score of 1 if a recent feedback prefix appears in the rationale, 0.5 if both feedback and a nonempty rationale are present without such a match, and 0 otherwise; it then averages over steps. Hint Helpfulness is the maximum of the fraction of hint-marked actions and half the Feedback Grounding score. Error-aware Replan currently uses the same action-history proxy as Adaptation Quality.

Track 3: closed-loop control. With Δ_t the simulator's recorded mastery gain at step t, the implementation computes

$$\begin{aligned}\text{MasteryGain} &= \max(0, M_T - \widehat{M}_0), \\ \text{RetentionGain} &= 0.5\max\left(0, \sum_t \Delta_t\right), \\ \text{LearnEff} &= \frac{\max(0, \sum_t \Delta_t)}{\max(1, T)}, \\ \text{SimExploit} &= \begin{cases} 1 - U/N_R, & N_R > 0, \\ 0, & N_R = 0, \end{cases}\end{aligned} \tag{S7}$$

Here U counts distinct selected resource identifiers and N_R counts resource-selection actions. Recovery Rate counts the fraction of post-incorrect-feedback opportunities followed by a designated recovery-oriented action (review, explanation, diagnosis quiz, easier problem, or plan update); it therefore measures a recovery response, not necessarily successful mastery restoration. Overload Rate uses the difficulty rule above. Action-type frequencies, average difficulty, target-concept hit rate, and the distinct-resource ratio are reported as diagnostic descriptors rather than uniformly optimized educational outcomes. In particular, Retention Gain is a simulator-derived proxy, not a delayed retention test.

F.5 Aggregate Educational Scores and Reporting Conventions

The public evaluator computes a raw educational score $S_{i,n}$ for episode n and clips it to $[0, 1]$. Under the paper's track numbering, its implemented expressions are

$$\begin{aligned} S_{1,n} &= 0.30\,\mathrm{SeqCons} + 0.25\,\mathrm{ResMatch} + 0.20\,\mathrm{ConsSat} \\ &\quad + 0.15\,(1 - \mathrm{PrereqViol}) + 0.10\,\mathrm{PathCoh}, \\ S_{2,n} &= 0.40\,\mathrm{MisAcc} + 0.30\,\mathrm{RemSuccess} + 0.20\,\mathrm{FeedGround} \\ &\quad + 0.10\,\mathrm{HintHelp}, \\ S_{3,n} &= 0.35\,\mathrm{MasteryGain} + 0.25\,\mathrm{RetentionGain} + 0.20\,\mathrm{LearnEff} \\ &\quad + 0.10(1 - \mathrm{DropoutRisk}) + 0.10(1 - \mathrm{SimExploit}). \end{aligned} \tag{S8}$$

The corresponding implemented episode score is $\mathrm{EduScore}_{i,n}^{\mathrm{impl}} = \mathrm{clip}_{[0,1]}(S_{i,n})$ and $\mathrm{Overall}_{i,n}^{\mathrm{impl}} = \mathrm{clip}_{[0,1]}[0.5\mathrm{Core}_n + 0.5\mathrm{EduScore}_{i,n}^{\mathrm{impl}}]$. Reported metrics are computed as arithmetic means of their *episode-level* values, not by first averaging component metrics and then clipping the result.

G Agent Configurations and Implementation Details

All nine evaluated variants use a common language model, episode observations, action schema, and track policy. Their difference is a method-inspired prompt, not a faithful implementation of the named original system. For agent family i and track j, the prompt is assembled as

$$\mathrm{Prompt}_{i,j} = \mathrm{Template}(\mathrm{Role}_i, \mathrm{Method}_i, \mathrm{Policy}_j, \mathrm{Context}_t). \tag{S9}$$

The agent returns one JSON action containing `action_type`, optional `resource_id`, `target_concepts`, `rationale`, `plan_update`, and `payload`. The context provides the learning goal, visible mastery, recent feedback, resource candidates, constraints, and current plan. A shared bridge normalizes actions and may replace invalid, repetitive, or policy-incompatible outputs with rule-based fallbacks; these repairs are part of the evaluated policy, not intrinsic capabilities of the original planning systems.

Variant	Method-specific prompt instruction
One-shot	Follow an initial fixed-horizon plan with minimal adaptation.
CoT	Reason privately about prerequisites, mastery, load, and drift.
Step-by-step	Select only the next valid action.
ReAct	Ground each action in current observation and feedback.
LLM+P-style	Represent state as symbolic facts and choose a planning operator.
LATS-style	Compare candidate actions through an internal search-style prompt.
Plan-and-Act-style	Separate planning from execution; replan when contradicted.
ReAcTree-style	Use hierarchical diagnostic, remediation, and practice branches.
HiAgent-style	Track goals, weak concepts, feedback, and constraints in prompt memory.

Table S1: Nine controlled, method-inspired prompt variants; no separate search engine, symbolic solver, or external memory runtime is implied.

Track-specific instructions prioritize prerequisite sequencing (Track 1), early diagnosis followed by remediation (Track 2), and readiness-aligned practice with limited repeated actions (Track 3). These instructions are shared by all variants within a track. Exact prompts and bridge behavior are available in the released implementation.

H Extended Experimental Results

This section provides fine-grained analyses of the nine prompt-based planning variants across the three EduPlanBench tracks, followed by a separate horizon-robustness evaluation. These results complement the main cross-track comparison by examining pedagogical process quality, diagnostic alignment, feedback-responsive behavior, and simulated learning outcomes.

H.1 Track 1: Prerequisite-Aware Planning

Track 1 evaluates whether agents can construct coherent learning paths while respecting prerequisite relations, resource constraints, and learner readiness. Table S2 reports Constraint Satisfaction and Difficulty Alignment.

Variant	ConsSat↑	DiffAlign↑
One-shot	.9211	.8304
CoT	.8984	.8141
Step-by-step	.9046	.8222
ReAct	.9163	.8304
LLM+P	.9065	.8264
LATS	.9043	.8183
Plan-and-Act	.9325	.8324
ReAcTree	.9418	.8327
HiAgent	.9384	.8420

Table S2: Track 1 path-quality diagnostics. ConsSat measures constraint satisfaction, while DiffAlign assesses resource difficulty relative to learner readiness.

Constraint Satisfaction ranges from 0.8984 to 0.9418, with ReAcTree achieving the highest value, while HiAgent obtains the highest Difficulty Alignment (0.8420). These results indicate relatively strong performance on structural feasibility and resource difficulty matching. However, the task-type analysis identifies adaptive replanning as more challenging than initial path construction. Thus, producing feasible learning paths does not necessarily imply effective revision under learner feedback or environmental changes.

H.2 Track 2: Misconception Diagnosis and Remediation

Track 2 examines whether agents can identify student errors and select interventions aligned with diagnostic evidence. Table S3 reports four complementary metrics: Misconception Accuracy (MisAcc), Remediation Match (RemMatch), Feedback Grounding (FeedGround), and Error-aware Replanning (ErrReplan).

Variant	MisAcc↑	RemMatch↑	FeedGround↑	ErrReplan↑
One-shot	.6011	.0584	.5243	.5574
CoT	.6544	.0620	.5259	.4314
Step-by-step	.6656	.0596	.5393	.4317
ReAct	.6633	.0632	.5302	.4320
LLM+P	.6400	.0610	.5575	.4293
LATS	.5478	.0654	.5316	.4374
Plan-and-Act	.7133	.0630	.5126	.4505
ReAcTree	.6033	.0591	.5176	.4303
HiAgent	.5333	.0650	.5030	.4399

Table S3: Track 2 diagnosis and remediation diagnostics. RemMatch is a reference-text overlap proxy, while ErrReplan measures observable replanning-related behavior rather than verified learning recovery.

Plan-and-Act achieves the highest MisAcc (0.7133), whereas LATS obtains the highest RemMatch (0.0654). LLM+P leads in FeedGround (0.5575), while One-shot records the highest ErrReplan (0.5574). The leadership differences indicate that diagnostic identification, reference alignment, feedback grounding, and replanning behavior capture distinct aspects of the interaction process.

Notably, RemMatch remains below 0.07 for all variants despite substantially higher MisAcc values. This reveals weak reference-text alignment in the evaluated interventions and shows that stronger diagnosis scores do not necessarily translate into higher remediation matching.

H.3 Track 3: KT-Based Closed-Loop Control

Track 3 evaluates feedback-conditioned intervention under benchmark-defined learner dynamics. Table S4 reports simulated Mastery Gain (Gain), Recovery Rate (Recovery), Dropout Risk (Dropout), and Simulator Exploitation (Exploit).

ReAcTree achieves the highest Gain (0.3849) and Recovery (0.4041), but also the highest Exploit value (0.5470), indicating more repetitive resource selection. HiAgent obtains the lowest simulated Dropout Risk (0.1011), while One-shot has the lowest Exploit score (0.4739).

These results reveal trade-offs between simulated learning improvement, recovery-oriented actions, and undesirable interaction patterns. In particular, frequent recovery actions do not necessarily produce effective

Variant	Gain↑	Recovery↑	Dropout↓	Exploit↓
One-shot	.3438	.3083	.1366	.4739
CoT	.3272	.3816	.1029	.5274
Step-by-step	.3218	.3641	.1184	.5308
ReAct	.3154	.3326	.1037	.5161
LLM+P	.3449	.3432	.1305	.4887
LATS	.3299	.2862	.1382	.5064
Plan-and-Act	.3311	.3294	.1046	.4999
ReAcTree	.3849	.4041	.1175	.5470
HiAgent	.3596	.3811	.1011	.5425

Table S4: Track 3 closed-loop diagnostics. Recovery measures recovery-oriented actions after incorrect feedback, while Dropout and Exploit represent simulator-defined risk and resource-repetition proxies.

mastery restoration, and stronger mastery gains may coexist with more repetitive behavior. No variant dominates all reported closed-loop criteria.

H.4 Horizon Robustness

We further evaluate whether agents benefit from additional intervention opportunities by varying the maximum planning horizon over $H \in \{10, 30, 50, 100\}$. The horizon specifies an intervention budget rather than a fixed number of executed actions. Let $S(H)$ denote the GSR under budget H. We report four complementary summaries:

$$\text{Early} = S(10), \qquad \Delta_H = S(100) - S(10), \tag{S10}$$

$$\text{Eff}_H = \frac{S(100) - S(10)}{90}, \quad \text{AUC-H} = \frac{1}{4} \sum_{H \in \{10,30,50,100\}} S(H). \tag{S11}$$

Early measures short-budget performance, Gain captures the endpoint improvement from increasing the available budget, and Eff normalizes this improvement by the additional budget allowance. AUC-H is the equally weighted mean GSR across the four evaluated budgets, rather than a continuous area integral.

Track 1. LATS and Plan-and-Act achieve the largest Horizon Gain (0.68), whereas CoT attains the highest AUC-H (0.755). Thus, the variants benefiting most from an expanded budget are not necessarily those maintaining the strongest average goal-level performance across budgets.

Track 2. Plan-and-Act achieves the highest AUC-H (0.875), while Step-by-step obtains the largest Horizon Gain (0.36). Several variants already exhibit relatively high Early performance, suggesting that their task-level success is less sensitive to additional intervention opportunities than in Track 1.

Track 3. One-shot and ReAcTree tie for the highest AUC-H (0.855), while Step-by-step achieves the largest Horizon Gain (0.50). These differences again separate strong initial or average performance from the incremental benefit of larger intervention budgets.

Interpretation. All evaluated variants exhibit positive endpoint GSR gains as the maximum intervention budget increases. However, these gains quantify improvements in task-aware success under different budget limits; they do not directly establish more accurate learner-state inference, effective feedback utilization, or successful recovery. Moreover, because GSR uses track-specific success criteria, cross-track differences in horizon metrics should not be interpreted as calibrated differences in task difficulty.

H.5 Summary of Findings

The extended results reveal complementary limitations across the three tracks. Track 1 shows that structurally feasible resource selection does not guarantee adaptive path revision. Track 2 exposes a gap between reference-based misconception identification and remediation alignment. Track 3 demonstrates trade-offs among simulated mastery improvement, recovery-oriented actions, and repetitive or risk-related behavior.

The horizon analysis further distinguishes short-budget effectiveness, endpoint gains, and average performance across intervention budgets. Taken together, these results support multidimensional trajectory evaluation: terminal task success alone cannot fully characterize the pedagogical coherence or feedback responsiveness of long-horizon educational agents. All learner-state changes and risk-related outcomes

Track	Variant	Early	Gain	Eff.	AUC-H
1	ReAct	.28	.56	.0062	.710
1	One-shot	.28	.56	.0062	.710
1	Step-by-step	.38	.50	.0056	.725
1	CoT	.42	.44	.0049	.755
1	LLM+P	.36	.42	.0047	.700
1	LATS	.26	.68	.0076	.730
1	Plan-and-Act	.18	.68	.0076	.660
1	ReAcTree	.28	.54	.0060	.705
1	HiAgent	.24	.60	.0067	.690
2	ReAct	.62	.34	.0038	.820
2	One-shot	.60	.34	.0038	.825
2	Step-by-step	.62	.36	.0040	.835
2	CoT	.62	.34	.0038	.815
2	LLM+P	.68	.30	.0033	.850
2	LATS	.64	.34	.0038	.835
2	Plan-and-Act	.68	.30	.0033	.875
2	ReAcTree	.66	.32	.0036	.830
2	HiAgent	.68	.30	.0033	.825
3	ReAct	.54	.42	.0047	.805
3	One-shot	.64	.34	.0038	.855
3	Step-by-step	.48	.50	.0056	.830
3	CoT	.54	.42	.0047	.820
3	LLM+P	.56	.40	.0044	.835
3	LATS	.54	.42	.0047	.805
3	Plan-and-Act	.54	.44	.0049	.825
3	ReAcTree	.54	.42	.0047	.855
3	HiAgent	.56	.42	.0047	.850

Table S5: Horizon robustness across tracks and prompt variants. Eff. measures gain per additional unit of available intervention budget, not per executed action.

remain defined by the benchmark simulator and should not be interpreted as direct evidence of real-world instructional effectiveness.

I Case Study

Twelve episodes include one success and three failures per track (Table S6). Mastery, reference misconceptions, and risk are evaluator-side information. Cases do not estimate failure frequencies or real-world instructional effectiveness.

Table S6: Representative successful and failure trajectories.

Case / Track	Agent	Outcome	Trajectory pattern / failure mode
T1-S / 1	HiAgent	GSR 1; PR 1	Prerequisite explanation followed by target explanation/exercises; efficient prerequisite-to-target path.
T1-F1 / 1	HiAgent	GSR 0; PR .067	24 explanation actions; locally relevant resources but path coherence collapses.
T1-F2 / 1	LLM+P	GSR 0; PR 0	Symbolic planning intent does not translate into an effective resource path.
T1-F3 / 1	One-shot	GSR 0; PR 0	Static sequence ignores an error streak and resource unavailability.
T2-S / 2	One-shot	GSR 1; PR 1	Diagnosis, remediation, and targeted practice form a feedback loop.
T2-F1 / 2	One-shot	GSR 0; PR 0	Misdiagnosis and fixed-plan drift after an incorrect exercise.
T2-F2 / 2	HiAgent	GSR 0; PR 0	Prerequisite forgetting followed by repeated explanations without recovery.
T2-F3 / 2	LATS	GSR 0; PR .171	Excessive explanation selection and insufficient assessment.
T3-S / 3	Plan-and-Act	GSR 1; PR 1	Incorrect target exercise, then review, then two correct exercises.
T3-F1 / 3	LATS	GSR 0; PR .124	Exercise-heavy policy delays diagnosis and prerequisite recovery.
T3-F2 / 3	ReAcTree	GSR 0; PR .124	Repeated target exercise raises dropout risk to .588.
T3-F3 / 3	ReAct	GSR 0; PR .150	Reactive loop repeatedly reuses one resource and fails to stabilize mastery.

Track 1: Prerequisite-Aware Planning. T1-S (`track1_005617`, HiAgent) progresses from neurophysiology prerequisite explanation to kinesthesia explanations and two correct exercises, raising mastery $0.200 \rightarrow 0.930$ in six steps.

T1-F1 (`track1_034462`, HiAgent) recommends 24 processor-scheduling prerequisite explanations, but path coherence is zero and mastery changes $0.200 \rightarrow 0.237$. T1-F2 (`track1_010033`, LLM+P) follows diagnosis with explanations and reviews for finite-dimensional vector spaces; mastery decreases $0.200 \rightarrow 0.157$ over 42 steps. T1-F3 (`track1_032596`, One-shot) retains a static sequence despite errors and resource unavailability, ending at mastery 0.172 after 42 steps. Local resource relevance does not ensure coherent, adaptive sequencing.

Track 2: Misconception Diagnosis and Remediation. T2-S (`track2_005766`, One-shot) addresses dividing 14 by meal count rather than rice quantity. Diagnosis, remediation and tutoring explanations, and a correct exercise raise mastery $0.450 \rightarrow 0.846$ in four steps.

T2-F1 (`track2_016927`, One-shot) alternates quadratic-graph explanations with mismatched geometry remediation after an incorrect exercise; mastery decreases $0.340 \rightarrow 0.229$ over 12 steps. T2-F2 (`track2_006530`, HiAgent) repeats explanations and easier problems after errors and prerequisite forgetting in order-of-operations; mastery decreases $0.240 \rightarrow 0.182$ over 14 steps. T2-F3 (`track2_003674`, LATS) overuses explanations with little assessment in factorisation, raising mastery $0.240 \rightarrow 0.336$. Plausible interventions coexist with remediation mismatch and insufficient assessment.

Track 3: KT-Based Closed-Loop Control. T3-S (`track3_005697`, Plan-and-Act) targets perimeter-strategy knowledge component 140. After a plan update and incorrect exercise, review and two correct exercises raise mastery $0.050 \rightarrow 0.868$ in five steps, exceeding the 0.750 threshold.

Under prerequisite-forgetting and forced-error perturbations, T3-F1 (`track3_006717`, LATS) uses 16 exercises, three diagnostics, and one review over 20 steps; late diagnosis leaves mastery at 0.137. T3-F2 (`track3_018024`, ReAcTree) repeats `q_2316` after incorrect feedback, ending at mastery 0.137 and dropout risk 0.588 after 20 steps. T3-F3 (`track3_001639`, ReAct) alternates diagnosis, review, and exercise but reuses `q_107`; late incorrect feedback leaves mastery at 0.155 after 20 steps. Reactive changes do not ensure recovery.

Successes connect prerequisites or diagnosis to intervention and assessment; failures exhibit drift, mismatch, under-assessment, or repetition. These observations describe actions and simulated outcomes without measuring internal belief updates.